Condensed Matter Physics

Assignment - 1

1. For the quadratic dispersion,
$$E = \frac{\hbar^2|k|^2}{2m} \tag{1}$$
find the density of states in d-dimensions and verify it with the known result in the 1-D & 3-D case.

2. Consider the exact dispersion relation of the 1-D harmonic lattice:
$$\omega(k) = \sqrt{\frac{4\alpha}{M}}\left|\sin\left(\frac{ka}{2}\right)\right| \tag{2}$$
where a is the lattice spacing, M is the mass of the atom, α is the inter-atomic force constant.

   (a) Find the density of states $g(E)$ and obtain the energy E_c up to which it is defined.

   (b) For a lattice with N sites, show that the number of states is given by
$$\int_0^{E_c} g(E)\, dE = N \tag{3}$$
.

3. The energy density of a gas of excitations with dispersion E($\mathbf{k}$) can be written as
$$u(T) = \frac{1}{(2\pi)^3}\int_0^{\infty} d^3k\; E(k)\, e^{-\beta E(k)}, \tag{4}$$
where $\beta = 1/(k_B T)$.

   (a) For the gapless excitation with the linear dispersion $E(\mathbf{k}) = \hbar v|k|$.
   Find the energy density and the low temperature behaviour of the specific heat.

   (b) Now, consider the gapped excitation, where the dispersion is given by $E(\mathbf{k}) = \hbar v|k| + \Delta$ (with $\Delta > 0$).
   Find the energy density and the low temperature behaviour of the specific heat.

4. To understand the origin of the Van der Waals interaction, consider the toy model of a one-dimensional atom where the electron with charge $-e$ is attached to a fixed nucleus of charge $+e$ by a spring with spring constant κ. The atom has a dipole moment $\vec{p} = -ex\hat{x}$ directed towards the nucleus, where x is the displacement of the electron, with respect to the nucleus, undergoing simple harmonic motion. Two such atoms are placed along a line at a distance R ($R >> x_1, x_2$). The instantaneous displacements of the two electrons from their respective nuclei are denoted by x_1 and x_2, as illustrated in Fig. 1. (Assume both atoms have the same spring constant κ).

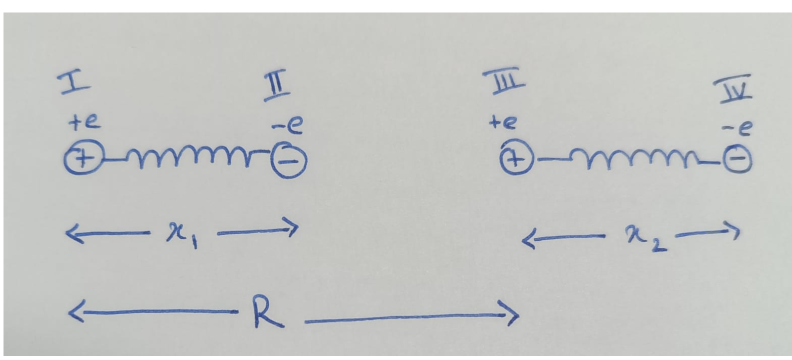

Figure 1: A system of two dipoles separated by a distance R

(a) Consider the system to be classical and write down the Hamiltonian of the system by considering the effective Coulomb interaction between the four charges. Show that to the leading order, the interaction energy goes as $E_{int} \sim R^{-3}$.

(b) Write down the equations of motion for the negatively charged particles and comment on the form of the equation in the limit $R \to \infty$.

(c) Perform the normal mode analysis for this system and find the frequencies of oscillations ω_+, ω_-.

(d) Considering these normal modes to be quantum mechanical harmonic oscillators, with ground state energy $\frac{\hbar\omega_+}{2}$ and $\frac{\hbar\omega_-}{2}$. Show that the leading order energy correction goes as $E \sim R^{-6}$

(e) Considering the initial Hamiltonian in (a) to be quantum mechanical, treat the Coulombic interaction as a small perturbation, and find the leading order correction to the ground state energy of the coupled system.